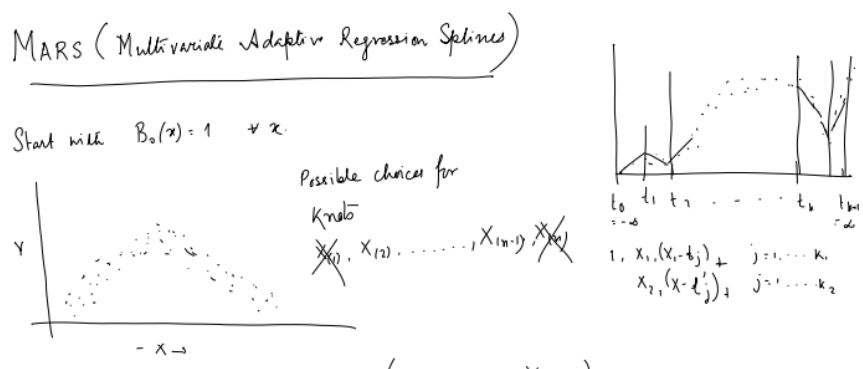


MARS (Multivariate Adaptive Regression Splines) and Classification / Bayes Decision Theory

Lecture Notes, Statistical Learning-I

1 MARS (Multivariate Adaptive Regression Splines)



Start with $B_0(x) = 1$ & x .

1.1 Possible choices for Knots

$$X_{(1)}, X_{(2)}, \dots, X_{(m-1)}, X_{(m)}$$

For each choice of the knot t ($t = X_{(1)}, \dots, X_{(m-1)}$) we consider a pair of functions

$$(x - t)_+ = \begin{cases} x - t & \text{if } x > t \\ 0 & \text{otherwise} \end{cases}$$

and

$$(t - x)_+ = \begin{cases} t - x & \text{if } t > x \\ 0 & \text{otherwise} \end{cases}$$

For this pair, we consider two new functions $B_0(x)(x - t)_+$ and $B_0(x)(t - x)_+$ and regress Y on $B_0(x)$, $B_0(x)(x - t)_+$, $B_0(x)(t - x)_+$.

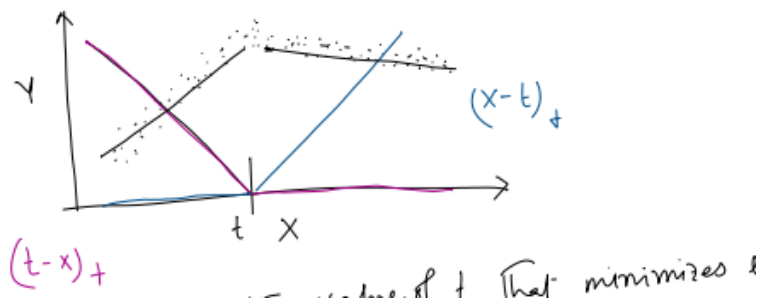
We minimize

$$\sum \left[Y_i - a_0 B_0(x_i) - a_1 B_0(x_i)(x_i - t)_+ - a_2 B_0(x_i)(t - x_i)_+ \right]^2$$

w.r.t. a_0, a_1 and a_2 and compute the corresponding RSS.

Finally, we shall choose the value of t that minimizes the residual SS. For instance if $t = X_{(k)}$ then we take $B_1(x) = B_0(x)(x - X_{(k)})_+$ & $B_2(x) = B_0(x)(X_{(k)} - x)_+$

$$B_0(x) \longrightarrow \{B_0(x), B_1(x), B_2(x)\}$$



2 Forward Step Continuation & Multivariate Case

Suppose that at some stage, we have m many basis functions

$$B_0(x), \dots, B_m(x).$$

At the next stage for all possible choices of t , we consider the pair $(x-t)_+$ & $(t-x)_+$ as before

$$\begin{array}{ll} B_0(x) & \\ B_1(x) & (x-t)_+ \\ \vdots & (t-x)_+ \\ B_m(x) & \end{array}$$

For each $j = 0, 1, \dots, m$ we consider two new functions $B_j(x)(x-t)_+$ and $B_j(x)(t-x)_+$ and compute the RSS for a model with $(m+1) + 2$ many basis functions

$$\{B_0, \dots, B_m, B_s(x)(t-x)_+, B_s(x)(x-t)_+\}$$

We do this for all choices of s and t and finally consider the one leading to the lowest RSS.

We proceed in this way and stop when the reduction in RSS is not significant.

2.1 Multivariate Case

For the multivariate case, the procedure remains almost the same.

There we start with $B_0(x) \sim 1$ as before.

At the next step, we have several choices for the pair:

$$\left. \begin{array}{l} (x_j - x_{(i)})_+ \\ (x_{(i)} - x_j)_+ \end{array} \right\} \begin{array}{l} \text{for } i = 2, \dots, n-1 \\ j = 1, 2, \dots, p \end{array}$$

For each of these pairs, we compute the RSS by minimizing

$$\sum \left[Y_n - a_0 B_0(x_n) - a_1 B_0(x_j)(x_{jn} - x_{j(i)})_+ - a_2 B_0(x_j)(x_{j(i)} - x_{jn})_+ \right]^2$$

w.r.t. a_0, a_1 and a_2 .

Finally we choose the pair leading to the lowest RSS.

Also, when we have $m+1$ many elements in the basis, we consider

$$\left. \begin{array}{l} B_s(x)(x_j - x_{j(i)})_+ \\ B_s(x)(x_{j(i)} - x_j)_+ \end{array} \right\}$$

as the possible pair of functions and compute the RSS for the model based on $(m+1) + 2$ many basis functions.

We do this for all choices of s ($s = 0, 1, \dots, m$), j ($j = 1, \dots, p$), and i ($i = 2, 3, \dots, n-1$) and finally select the one which gives the lowest RSS.

We proceed this way until the reduction in RSS after adding two new functions is negligible.

Interaction / Degree control

Sometimes, we put an upper bound on the order of interaction (or the degree of the polynomial).

$$\begin{array}{c}
 \text{1} \\
 \left| \begin{array}{c} \text{1} \\ (x_1 - 3)_+ \\ (3 - x_1)_+ \end{array} \right| \left| \begin{array}{c} \text{1} \\ (x_1 - 3)_+ \\ (3 - x_1)_+ \\ (x_1 - 7)_+ \\ (7 - x_1)_+ \end{array} \right| \left| \begin{array}{c} \text{1} \\ (x_1 - 3)_+ \\ (3 - x_1)_+ \\ (x_1 - 7)_+ \\ (7 - x_1)_+ \\ (x_1 - 3)_+ (x_3 - 11)_+ \\ (x_1 - 3)_+ (11 - x_3)_+ \end{array} \right|
 \end{array}$$

3 Backward Deletion & Classification / Regression Setup

After choosing the set of basis functions among many models we often use the backward deletion method for getting parsimonious models. During backward deletion, we get a nested sequence of models and one of them is finally chosen using an appropriate criterion.

Classification	Regression $(X_1, y_1), \dots, (X_n, y_n)$
$(X_1, y_1), \dots, (X_n, y_n)$	
$c_1, c_2, \dots, c_n \in \{1, 2, \dots, J\}$	$y_1, \dots, y_n \in \mathbb{R}$
A, B, ... T	real-valued
$\delta : \mathbb{R}^d \rightarrow \{1, 2, \dots, J\}$	Aim: $f : \mathbb{R}^d \rightarrow \mathbb{R}$
$x \in \mathbb{R}^d, \delta(x) \in \{1, \dots, J\}$	$x \in \mathbb{R}^d, f(x) \in \mathbb{R}$
class label	response value
$L(c, \delta)$: loss function	$L(y, f)$: loss function
$R(f) = E[L(c, \delta(x))]$	$Risk(f) = E[L(y, f(x))]$
e.g. $E[1_{\{c \neq \delta(x)\}}]$	e.g. $E(Y - f(x))^2$ sq. error loss
$= P(\delta(x) \neq c)$	$E Y - f(x) $ absolute error loss

3.1 Parametric Classifier (Two-class)

Consider a parametric classifier for a two-class problem

$$(x_1, 1), (x_2, 1), (x_3, 2), \dots, (x_n, 2)$$

$$c \in \{1, 2\}, \quad \delta \in \{1, 2\}$$

$$\delta(x) = \begin{cases} 1 & \text{if } \alpha + \beta x > 0 \\ 2 & \text{if } \alpha + \beta x < 0 \end{cases}$$

(consider a problem with one feature variable)

$$Risk = E[1_{\{c \neq \delta(x)\}}]$$

$$c \begin{cases} 1 & f_1 \leftarrow \text{density for class 1} \\ 2 & f_2 \leftarrow \text{density for class 2} \end{cases}$$

$$Risk(\delta) = P(\delta(x) \neq c) = P(c=1)P(\delta(x) \neq 1 | c=1) + P(c=2)P(\delta(x) \neq 2 | c=2)$$

3.2 Parametric Regression

Consider a parametric regression problem where we have only one predictor variable X and model we consider

$$f(x) = \alpha + \beta x$$

$$Risk = E[Y - f(x)]^2 = E[Y - \alpha - \beta x]^2$$

In practice, we minimize an empirical analog of the risk

$$\frac{1}{n} \sum (y_i - \alpha - \beta x_i)^2 \quad \text{or minimize} \quad \sum (y_i - \alpha - \beta x_i)^2$$

Handwritten notes and diagrams illustrating the risk function and its empirical version.

Risk function derivation:

$$\begin{aligned} Risk(\delta) &= P(\delta(x) \neq c) \\ &= P(c=1)P(\delta(x) \neq 1 | c=1) + P(c=2)P(\delta(x) \neq 2 | c=2) \\ &= \pi_1 \int_{\delta(x) \neq 1} f_1(x) dx + \pi_2 \int_{\delta(x) \neq 2} f_2(x) dx \end{aligned}$$

Bayesian Rule:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

$$P(A \cap B) = P(B)P(A|B)$$

$$P(A \cap B^c) = P(B^c)P(A|B^c)$$

Empirical version of the risk:

$$\frac{n_2}{n} \sum_{i=1}^{n_2} 1\{\alpha + \beta x_{2i} > 0\} + \frac{n_1}{n} \sum_{i=1}^{n_1} 1\{\alpha + \beta x_{1i} \leq 0\}$$

Diagram: A Venn diagram showing two overlapping circles labeled n_1 and n_2 . The region n_1 is labeled "Class 1" and the region n_2 is labeled "Class 2". The intersection of the two circles is labeled n_{12} . The regions outside the intersection are labeled $n_1 - n_{12}$ and $n_2 - n_{12}$.

4 Risk Expressions and Empirical Risk

$$= \pi_1 \int_{\delta(x) \neq 1} f_1(x) dx + \pi_2 \int_{\delta(x) \neq 2} f_2(x) dx$$

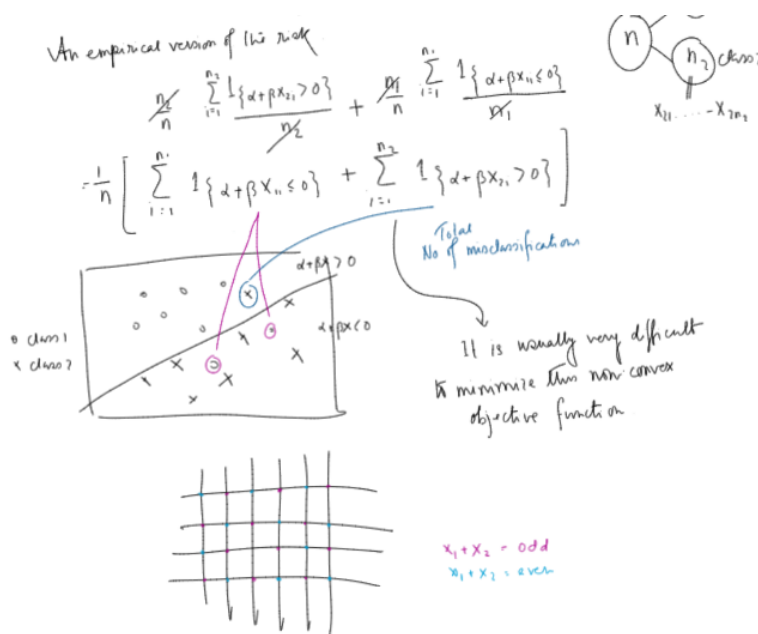
$$Risk(\delta) = P(\delta(x) \neq c) = P(\delta(x) = 1, c=2) + P(\delta(x) = 2, c=1) = P(\alpha + \beta x > 0, c=2) + P(\alpha + \beta x \leq 0, c=1)$$

An empirical version of the risk:

$$\frac{n_2}{n} \sum_{i=1}^{n_2} 1\{\alpha + \beta x_{2i} > 0\} + \frac{n_1}{n} \sum_{i=1}^{n_1} 1\{\alpha + \beta x_{1i} \leq 0\} = \frac{1}{n} \left[\sum_{i=1}^{n_1} 1\{\alpha + \beta x_{1i} \leq 0\} + \sum_{i=1}^{n_2} 1\{\alpha + \beta x_{2i} > 0\} \right]$$

Total number of misclassifications.

It is usually very difficult to minimize this non-convex objective function.



5 Two-Class Classification Risk (0-1 Loss)

In a two-class classification problem, if we use the 0-1 loss function, the risk (i.e. the expected loss) of a classifier δ is given by

$$P(\delta(X) \neq c).$$

If π_1 and π_2 are the prior probabilities of the two classes and f_1 and f_2 are their density functions, this can be expressed as

$$\begin{aligned} P(\delta(X) \neq c) &= P(\delta(X) = 2, c = 1) + P(\delta(X) = 1, c = 2) \\ &= P(c = 1)P(\delta(X) = 2 \mid c = 1) + P(c = 2)P(\delta(X) = 1 \mid c = 2) \\ &= \pi_1 \int_{x:\delta(x)=2} f_1(x) dx + \pi_2 \int_{x:\delta(x)=1} f_2(x) dx. \end{aligned}$$

An empirical version of this risk is given by

$$\frac{n_1}{n} \cdot \frac{1}{n_1} \sum_{i=1}^{n_1} I\{\delta(x_{i1}) = 2\} + \frac{n_2}{n} \cdot \frac{1}{n_2} \sum_{i=1}^{n_2} I\{\delta(x_{i2}) = 1\} = \frac{1}{n} \left[\sum_{i=1}^{n_1} I\{\delta(x_{i1}) = 2\} + \sum_{i=1}^{n_2} I\{\delta(x_{i2}) = 1\} \right],$$

the total proportion of misclassification.

Even for a simple classification method (eg linear classification) it may not be computationally feasible to minimize the empirical version of the risk over a collection of classifiers in that class. This is mainly because this empirical version is not a convex function.

Here one can think of minimizing

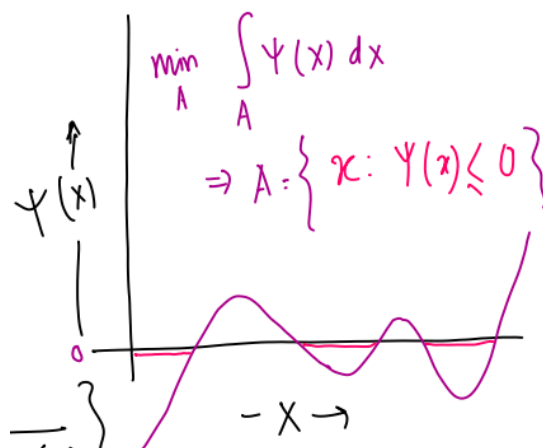
$$\begin{aligned} &\pi_1 \int_{x:\delta(x)=2} f_1(x) dx + \pi_2 \int_{x:\delta(x)=1} f_2(x) dx \quad \text{over } \delta. \\ &= \pi_1 \int_{R_2(\delta)} f_1(x) dx + \pi_2 \int_{R_1(\delta)} f_2(x) dx = \pi_1 \left\{ 1 - \int_{R_1(\delta)} f_1(x) dx \right\} + \pi_2 \int_{R_1(\delta)} f_2(x) dx \end{aligned}$$

$$= \pi_1 + \int_{R_1(\delta)} [\pi_2 f_2(x) - \pi_1 f_1(x)] dx.$$

This risk is minimized when

$$R_1(\delta) = \{x : \pi_1 f_1(x) > \pi_2 f_2(x)\} = \left\{x : \frac{\pi_1 f_1(x)}{\pi_1 f_1(x) + \pi_2 f_2(x)} > \frac{\pi_2 f_2(x)}{\pi_1 f_1(x) + \pi_2 f_2(x)}\right\} = \{x : p(1 | x) > p(2 | x)\}$$

(posterior probability of class 1).



Example, Defective Items

Company → Machine I → Machine II

- Machine I manufactures 60% of its products.
- 20% of the items produced by Machine I are defective.
- 30% of the items produced by Machine II are defective.

Purchased a pack of 20 items (5 defective, 15 non-defective) all handled by one machine.

Which machine is more likely to be the manufacturer?

6 Bayes Classifier – Machine Example & Lifetime Example

6.1 Defective Pack Example

Let X = number of defectives in a pack of 20.

$$P(M_1) = 0.6, \quad P(M_2) = 0.4 \quad (\pi_1, \pi_2)$$

$$X | M_1 \sim \text{Bin}(20, 0.2), \quad f_1(x) = \binom{20}{x} (0.2)^x (0.8)^{20-x}$$

$$X | M_2 \sim \text{Bin}(20, 0.3), \quad f_2(x) = \binom{20}{x} (0.3)^x (0.7)^{20-x}$$

$$P(M_1 | X = 5) = \frac{P(M_1)P(X = 5 | M_1)}{P(X = 5)} = \frac{\pi_1 f_1(5)}{\pi_1 f_1(5) + \pi_2 f_2(5)}$$

We assign the pack (here class $X = 5$) to the first class (M_1) if

$$\pi_1 f_1(5) > \pi_2 f_2(5).$$

Had x been the number of defectives, it would have been classified to M_1 if

$$\pi_1 f_1(x) > \pi_2 f_2(x).$$

Bayes classifier / Optimal classifier / Oracle classifier

$$\delta_{\text{Bayes}}(x) = \arg \max_{i=1,2} \pi_i f_i(x)$$

6.2 Lifetime Bulbs Example

- Company: 70% Type A – lifetime follows Exp with mean 2000 hrs.
- 30% Type B – lifetime follows Exp with mean 3000 hrs.

Purchased two bulbs of the same type. Observed their life times: 2500 hrs & 2700 hrs respectively.

$X = (X_1, X_2)$: lifetimes of two bulbs (feature vector)

$$f_1(x_1, x_2) = \frac{1}{2000} e^{-x_1/2000} \cdot \frac{1}{2000} e^{-x_2/2000} = \left(\frac{1}{2000}\right)^2 e^{-\frac{1}{2000}(x_1+x_2)}$$

$$f_2(x_1, x_2) = \left(\frac{1}{3000}\right)^2 e^{-\frac{1}{3000}(x_1+x_2)}$$

$$\pi_1 = 0.7, \quad \pi_2 = 0.3$$

$$\pi_1 f_1(x_1, x_2) > \pi_2 f_2(x_1, x_2)$$

leads to a linear decision boundary in the $(x_1 + x_2)$ plane.

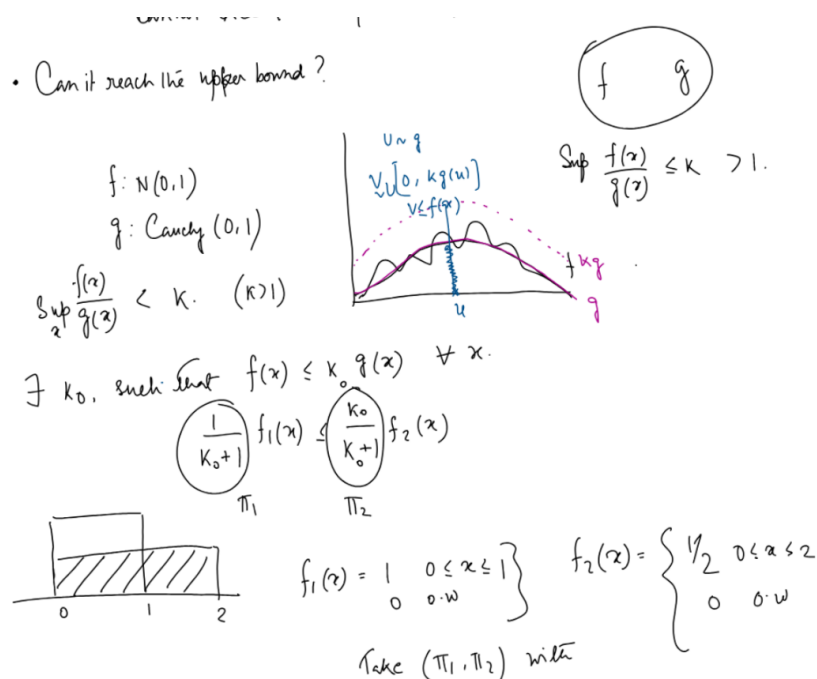
Bayes rule:

$$\delta(x) = \arg \max_i (\pi_i f_i(x)) = \arg \max_i p(i | x) = \arg \max_i \frac{\pi_i f_i(x)}{\sum_j \pi_j f_j(x)}$$

In a two-class classification problem, the Bayes risk (i.e. risk of the Bayes classifier) cannot exceed $\min\{\pi_1, \pi_2\}$. Can it reach the upper bound?

7 Properties of the Bayes Rule

7.1 Boundedness of Density Ratios



$$q \sim \text{Cauchy}(0,1)$$

Suppose $\frac{f(x)}{q(x)} < K$ ($K > 1$). There exists K_0 such that $f(x) \leq K_0 q(x) \quad \forall x$. Then

$$\frac{1}{K_0+1} f_1(x) \leq \frac{K_0}{K_0+1} f_2(x)$$

(with appropriate priors π_1, π_2).

7.2 Example with Uniform Densities

$$f_1(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad f_2(x) = \begin{cases} 1/2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Take (π_1, π_2) with $\pi_2 > 2/3$ (i.e. $\pi_1 < 1/3$).

7.3 Is the Bayes Rule Unique?

$$\delta(x) = \begin{cases} 1 & \text{if } \pi_1 f_1(x) \geq \pi_2 f_2(x) \\ 2 & \text{if } \pi_1 f_1(x) < \pi_2 f_2(x) \end{cases}$$

With $\pi_1 = 1/3$, $\pi_2 = 2/3$:

Region	$\pi_1 f_1(x)$	$\pi_2 f_2(x)$
(0, 1)	1/3	1/3
(1, 2)	0	1/3

Possible rules:

$$\delta_1(x) = \begin{cases} 1 & x \leq 1 \\ 2 & x > 1 \end{cases}$$

$$\delta_2(x) = \begin{cases} 1 & x \leq 0.5 \\ 2 & x > 0.5 \end{cases}$$

$$\delta_3(x) = \begin{cases} 1 & x \leq 0.1 \\ 2 & x > 0.1 \end{cases}$$

$$P(0.5 < X < 1) =$$

$$P(\delta_1(X) \neq \delta_2(X)) > 0$$

Hence the Bayes classifier is **not unique**.

7.4 Gaussian Case (Equal Covariance)

$$f_1 \sim N(\mu_1, \Sigma), \quad f_2 \sim N(\mu_2, \Sigma)$$

$$\pi_1 f_1(x) > \pi_2 f_2(x) \iff \log \pi_1 - \frac{1}{2}(x - \mu_1)^\top \Sigma^{-1}(x - \mu_1) > \log \pi_2 - \frac{1}{2}(x - \mu_2)^\top \Sigma^{-1}(x - \mu_2)$$

Property

Show that when the priors are equal, i.e. $\pi_1 = \pi_2 = 1/2$, the risk of this classifier is a decreasing function of the Mahalanobis distance

$$\sqrt{(\mu_1 - \mu_2)^\top \Sigma^{-1}(\mu_1 - \mu_2)}.$$

End of notes